

Permutation Subsequence

Time limit: 1.00 s

Memory limit: 512 MB

Given two arrays which are permutations, find their longest common subsequence.

A subsequence is a sequence of array elements from left to right that can contain gaps. A common subsequence is a subsequence that appears in both arrays.

Input

The first line has two integers n and m : the sizes of the arrays.

The second line has n integers $a_1, a_2, \dots, a_n$: the contents of the first array.

The third line has m integers $b_1, b_2, \dots, b_m$: the contents of the second array.

Output

First print the length of the longest common subsequence.

After that, an example of such a sequence. If there are several solutions, you can print any of them.

Constraints

- $1 \leq n, m \leq 2 \cdot 10^5$
- $1 \leq a_i \leq n$
- $1 \leq b_i \leq m$

Example

Input	Output
8 6 3 1 2 8 5 7 6 4 6 5 1 2 3 4	3 1 2 4